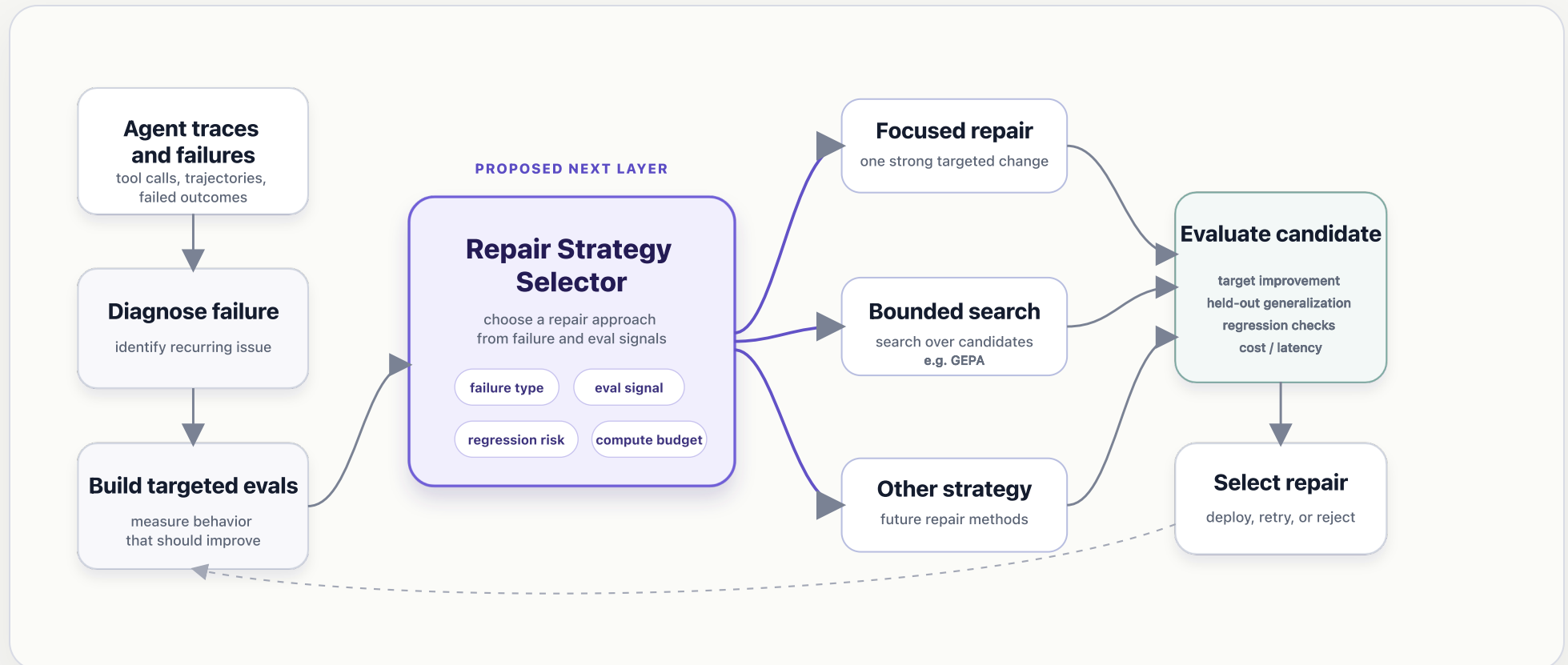


Choosing How to Repair Failing Agents

A repair-strategy selection layer for agent improvement loops

For systems that diagnose failures and build targeted evals, what repair strategy should run next? Mettle tests how different repair approaches trade off generalization, regression risk, and search cost.



What Mettle tested

Mettle compares repair approaches for a failing stateful tool-calling agent using separate optimization, untouched held-out, and regression evaluations. In the completed experiment, more search did not always produce the best generalizing repair, while different strategies showed different regression tradeoffs.

Possible next step

Choose the repair method dynamically based on the diagnosed failure, expected improvement, regression risk, and available compute, rather than defaulting to the same repair process for every issue.

Goal: spend search budget where it is most likely to help.